

Coherent perfect absorption by a dipolar scatterer coupled to a PEC

This set of notes concerns the paper Coherent perfect absorption by a single nanoparticle (<https://www.spiedigitallibrary.org/conference-proceedings-of-spie/12131/121310P/Coherent-perfect-absorption-by-a-single-nanoparticle/10.1117/12.2621039.short>) by **Proskurin et al.** We note that, in the below, all vectors are written in the $(\mathbf{e}_\rho, \mathbf{e}_\theta, \mathbf{e}_z)$ cylindrical basis (unless otherwise specified). We use θ instead of ϕ to denote the cylindrical azimuthal angle. In addition, unless otherwise specified, all notation is the same as the paper by **Proskurin et al.**

Section 1: The incident CPA field:

The incident field, which we tune later to achieve CPA, must be composed of **TM** and **TE** vector cylindrical waves. In the below, we take a linear combination of such waves with angular momentum number, $m = 1$. In order to ensure that the quantities $E_{1,2}(k_\rho)$ used below have units of electric field, we normalize each curl with a transverse momentum (i.e. $\nabla \times \rightarrow k_\rho^{-1} \nabla \times$, as well as $\nabla \times \nabla \times \rightarrow k_\rho^{-2} \nabla \times \nabla \times$). See also **Page 195** of the book by **Bohren and Huffman** for a general explanation of cylindrical wave solutions to the vector Helmholtz equation. Our incident field, $\mathbf{E}^{\text{inc}}$ is thus given as follows:

$$\begin{aligned} \mathbf{E}_{k_\rho}^{\text{inc}}(\rho, \theta, z) &= \frac{E_2(k_\rho)}{k_\rho} \nabla \times \left[\mathbf{e}_z J_1(k_\rho \rho) e^{-i\theta} e^{-ik_z z} \right] + \frac{E_1(k_\rho)}{k_\rho^2} \nabla \times \nabla \times \left[\mathbf{e}_z J_1(k_\rho \rho) e^{-i\theta} e^{-ik_z z} \right] \\ &= \tilde{E}_2(k_\rho, z, \theta) \begin{bmatrix} -\frac{i}{k_\rho \rho} J_1(k_\rho \rho) \\ -J_1'(k_\rho \rho) \\ 0 \end{bmatrix} + \tilde{E}_1(k_\rho, z, \theta) \begin{bmatrix} -i(k_z/k_\rho) J_1'(k_\rho \rho) \\ -(k_z/k_\rho)(1/k_\rho \rho) J_1(k_\rho \rho) \\ (1/k_\rho \rho) \left[-\partial_\rho(\rho J_1'(k_\rho \rho)) + (1/k_\rho \rho) J_1(k_\rho \rho) \right] \end{bmatrix} \end{aligned}$$

where we introduced $\tilde{E}_{1,2}(k_\rho, z, \theta) \equiv E_{1,2}(k_\rho) e^{-ik_z z - i\theta}$, so that we could omit the lengthy factor of $e^{-i\theta} e^{-ik_z z}$ in the last equality. Note also that we

have written the incident field for each transverse momentum, k_ρ , and that the total incident field is given by an integral over k_ρ of the above. At this point, we note the following (defining $x = k_\rho \rho$, $\partial_\rho = k_\rho \partial_x$):

$$\frac{1}{k_\rho \rho} \left[-\partial_\rho(\rho J_1'(k_\rho \rho)) + (1/k_\rho \rho) J_1(k_\rho \rho) \right] = \frac{1}{x} \left[-\partial_x(x J_1'(x)) + (1/x) J_1(x) \right] = -J_1''(x) - \frac{1}{x} J_1'(x) + \frac{1}{x^2} J_1(x) = -\frac{1}{x^2} \left[x^2 J_1''(x) + x J_1'(x) - J_1(x) \right] = J_1(x),$$

where the last relation follows from Bessel's equation for $m = 1$. Thus, we may write the incident field as follows:

$$\mathbf{E}_{k_\rho}^{\text{inc}}(\rho, \theta, z) = \tilde{E}_2(k_\rho, z, \theta) \begin{bmatrix} -\frac{i}{k_\rho \rho} J_1(k_\rho \rho) \\ -J_1'(k_\rho \rho) \\ 0 \end{bmatrix} + \tilde{E}_1(k_\rho, z, \theta) \begin{bmatrix} -i(k_z/k_\rho) J_1'(k_\rho \rho) \\ -(k_z/k_\rho)(1/k_\rho \rho) J_1(k_\rho \rho) \\ J_1(k_\rho \rho) \end{bmatrix}$$

which is **Equation 1** of the paper (after we integrate over the transverse momentum, k_ρ , of course).

Section 2: The reflected field associated with the incident CPA field

The reflected field is uniquely determined by the **PEC boundary conditions**, which require the tangential components of the electric field and the normal components of the magnetic field to vanish at the surface of the PEC. We put the scatterer at $z = 0$, $\rho = 0$ and the PEC boundary at the plane defined by $z = -h$. Thus, the incident field at the PEC boundary is given by:

$$\mathbf{E}_{k_\rho}^{\text{inc}}(\rho, \theta, z = -h) = E_2(k_\rho) \begin{bmatrix} -\frac{i}{k_\rho \rho} J_1(k_\rho \rho) \\ -J_1'(k_\rho \rho) \\ 0 \end{bmatrix} e^{-i\theta} e^{ik_z h} + E_1(k_\rho) \begin{bmatrix} -i(k_z/k_\rho) J_1'(k_\rho \rho) \\ -(k_z/k_\rho)(1/k_\rho \rho) J_1(k_\rho \rho) \\ J_1(k_\rho \rho) \end{bmatrix} e^{-i\theta} e^{ik_z h},$$

where we just evaluated **Equation 1** of the paper on the plane $z = -h$.

The $\mathbf{e}_\rho, \mathbf{e}_\theta$ components of the above must be cancelled out exactly.

Therefore, the reflected field must be as follows:

$$\mathbf{E}_{k_\rho}^{\text{ref}}(\rho, \theta, z) = E_2(k_\rho) \begin{bmatrix} \frac{i}{k_\rho \rho} J_1(k_\rho \rho) \\ J_1'(k_\rho \rho) \\ 0 \end{bmatrix} e^{-i\theta} e^{ik_z z} e^{2ik_z h} +$$

$$E_1(k_\rho) \begin{bmatrix} i(k_z/k_\rho) J_1'(k_\rho \rho) \\ (k_z/k_\rho)(1/k_\rho \rho) J_1(k_\rho \rho) \\ J_1(k_\rho \rho) \end{bmatrix} e^{-i\theta} e^{ik_z z} e^{2ik_z h},$$

which is **Equation 2** of the paper. We note that the total field, which is defined as the background field, takes the following form:

$$\mathbf{E}_{k_\rho}^{\text{bg}}(\rho, \theta, z) = \frac{E_2(k_\rho)}{k_\rho} \nabla \times [\mathbf{e}_z J_1(k_\rho \rho) e^{-i\theta} [e^{-ik_z z} - e^{2ik_z h} e^{ik_z z}]] +$$

$$\frac{E_1(k_\rho)}{k_\rho^2} \nabla \times \nabla \times [\mathbf{e}_z J_1(k_\rho \rho) e^{-i\theta} [e^{-ik_z z} + e^{2ik_z h} e^{ik_z z}]]$$

Section 2.5: Background electric field

The background electric field is defined as the field that would exist in the absence of the dipolar scatterer. I.e. it is given by $\mathbf{E}^{\text{inc}} + \mathbf{E}^{\text{ref}}$. We are concerned with calculating this background field at the position of the dipole, since the field at the location of the dipole determines the dipole moment via the polarizability tensor. As such, we are only concerned with the $\rho \rightarrow 0$ behavior of the background field. Thus, we may make the approximations $J_1(k_\rho \rho)/(k_\rho \rho) \approx 1/2$, and $J_1'(k_\rho \rho) \approx 1/2$. With these two approximations, we obtain:

$$\mathbf{E}_{k_\rho}^{\text{bg}}(\rho \approx 0, \theta, z = 0) \approx E_2(k_\rho) \begin{bmatrix} -\frac{i}{2}(1 - e^{2ik_z h}) \\ -\frac{1}{2}(1 - e^{2ik_z h}) \\ 0 \end{bmatrix} e^{-i\theta} - \frac{E_1(k_\rho)k_z}{k_\rho} \begin{bmatrix} \frac{i}{2}(1 - e^{2ik_z h}) \\ \frac{1}{2}(1 - e^{2ik_z h}) \\ 0 \end{bmatrix}$$

Thus, if the polarizability is isotropic, we have $\mathbf{p} \propto (1, -i, 0)$ (in the $(\mathbf{e}_\rho, \mathbf{e}_\theta, \mathbf{e}_z)$ basis). Note that this result (i.e. the overall ratio between the $\mathbf{e}_\rho$ and $\mathbf{e}_\theta$ components of the background electric field, evaluated at the dipole location), is independent of the integral we must take over the transverse momentum, k_ρ . Furthermore, note that:

$$e^{-i\theta}(\mathbf{e}_\rho - i\mathbf{e}_\theta) = e^{-i\theta} \left[\cos(\theta)\mathbf{e}_x + \sin(\theta)\mathbf{e}_y - i \left[-\sin(\theta)\mathbf{e}_x + \cos(\theta)\mathbf{e}_y \right] \right] = \mathbf{e}_x - i\mathbf{e}_y$$

This means that the electric field and the dipole moment have a well defined direction at the origin.

Section 2.6: Sommerfeld identity

We do not bother rederiving the Sommerfeld identity since we already did so in this set of notes: [Sommerfeld representation of the electromagnetic Green's function](https://hackmd.io/@aligho/SJmejWQ2Zg) (<https://hackmd.io/@aligho/SJmejWQ2Zg>). We just reiterate it as follows (and in the notation of the paper):

$$\frac{e^{ikr}}{4\pi r} = \frac{i}{4\pi} \int_0^\infty \frac{J_0(k_\rho \rho) e^{i|z|\sqrt{k_0^2 - k_\rho^2}}}{\sqrt{k_0^2 - k_\rho^2}} k_\rho dk_\rho,$$

Section 2.7: Form of the Green's function

We also don't bother rederiving the Dyadic Green's function and refer the reader to this set of notes: [Coupled dipole model for coherent perfect absorption](https://hackmd.io/@aligho/HJXva_Dv-e) (https://hackmd.io/@aligho/HJXva_Dv-e). We simply reiterate the main relevant result of that set of notes below:

$$\mathbf{E}_{\text{dipole}}(\mathbf{r}) = \frac{1}{\epsilon_0} \left[\frac{\omega^2}{c^2} + \nabla \nabla \right] G(\mathbf{r}, \mathbf{r}') \mathbf{p}(\mathbf{r}')$$

Section 3: Scattered field of the dipole

From the above, we see that the scattered field of the dipole will be given as follows:

$$\mathbf{E}^{\text{scat}}(\rho, \theta, z) = \frac{ip}{4\pi\epsilon_0} \left[k_0^2 + \nabla \nabla \cdot \right] \int_0^\infty \frac{J_0(k_\rho \rho) e^{ik_z|z|} e^{-i\theta}}{k_z} (\mathbf{e}_\rho - i\mathbf{e}_\theta) k_\rho dk_\rho,$$

where the dipole is $\mathbf{p} = p(\mathbf{e}_\rho - i\mathbf{e}_\theta) e^{-i\theta} = p(\mathbf{e}_x - i\mathbf{e}_y)$. We first evaluate the divergence: $\nabla \cdot \left(J_0(k_\rho \rho) e^{ik_z|z|} e^{-i\theta} (\mathbf{e}_\rho - i\mathbf{e}_\theta) \right)$, which we must determine in order to evaluate the second term in the dipole's electric field:

$$\begin{aligned} \nabla \cdot \left(J_0(k_\rho \rho) e^{ik_z|z|} e^{-i\theta} (\mathbf{e}_\rho - i\mathbf{e}_\theta) \right) &= \frac{1}{\rho} \left[\partial_\rho (\rho J_0(k_\rho \rho)) - J_0(k_\rho \rho) \right] e^{ik_z|z|} e^{-i\theta} = \\ &= -k_\rho J_1(k_\rho \rho) e^{ik_z|z|} e^{-i\theta} \end{aligned}$$

Furthermore, the gradient of the above is given by:

$$\nabla \left[-k_\rho J_1(k_\rho \rho) e^{ik_z|z|} e^{-i\theta} \right] = \left[-k_\rho^2 J_1'(k_\rho \rho) \mathbf{e}_\rho + \frac{ik_\rho}{\rho} J_1(k_\rho \rho) \mathbf{e}_\theta \mp ik_z k_\rho J_1(k_\rho \rho) \right]$$

Finally, we evaluate the quantity $k_0^2 J_0(k_\rho \rho) e^{-i\theta - ik_z|z|} (\mathbf{e}_\rho - i\mathbf{e}_\theta)$ (by which we mean we re-express it slightly in order to write it in terms of the Bessel function of order unity and its derivative):

$$\begin{aligned} k_0^2 J_0(k_\rho \rho) e^{-i\theta - ik_z|z|} (\mathbf{e}_\rho - i\mathbf{e}_\theta) &= \left[\frac{k_0^2}{k_\rho \rho} J_1(k_\rho \rho) + k_0^2 J_1'(k_\rho \rho) \right] (\mathbf{e}_\rho - i\mathbf{e}_\theta) = \\ &= \left[\frac{k_0^2}{k_\rho \rho} J_1(k_\rho \rho) \mathbf{e}_\rho - ik_0^2 J_1'(k_\rho \rho) \mathbf{e}_\theta \right] + \left[k_0^2 J_1'(k_\rho \rho) \mathbf{e}_\rho - \frac{ik_0^2}{k_\rho \rho} J_1(k_\rho \rho) \mathbf{e}_\theta \right], \end{aligned}$$

where in the last equality we judiciously separated the resultant into a sum of two expressions, which will allow us to write our final answer in terms of the quantities $\mathbf{v}_{1,2}^\pm$ defined in the paper. Putting everything together, we obtain the following:

$$\begin{aligned}
& \frac{ip}{4\pi\epsilon_0} \left[k_0^2 + \nabla \nabla \cdot \right] \frac{J_0(k_\rho \rho) e^{ik_z|z|} e^{-i\theta}}{k_z} (\mathbf{e}_\rho - i\mathbf{e}_\theta) k_\rho = \\
& \frac{ip}{4\pi\epsilon_0} \frac{k_\rho}{k_z} \left[\left[\frac{k_0^2}{k_\rho \rho} J_1(k_\rho \rho) \mathbf{e}_\rho - ik_0^2 J_1'(k_\rho \rho) \mathbf{e}_\theta \right] + \left[k_0^2 J_1'(k_\rho \rho) \mathbf{e}_\rho - \frac{ik_0^2}{k_\rho \rho} J_1(k_\rho \rho) \mathbf{e}_\theta \right] \right. \\
& \left. \left[-k_\rho^2 J_1'(k_\rho \rho) \mathbf{e}_\rho + \frac{ik_\rho}{\rho} J_1(k_\rho \rho) \mathbf{e}_\theta \mp ik_z k_\rho J_1(k_\rho \rho) \mathbf{e}_z \right] \right] e^{-i\theta} e^{ik_z|z|} = \\
& \frac{ipk_\rho}{4\pi\epsilon_0 k_z} \left[k_0^2 \left[\frac{J_1(k_\rho \rho)}{k_\rho \rho} \mathbf{e}_\rho - iJ_1'(k_\rho \rho) \mathbf{e}_\theta \right] + k_z^2 \left[J_1'(k_\rho \rho) \mathbf{e}_\rho - \frac{i}{k_\rho \rho} J_1(k_\rho \rho) \mathbf{e}_\theta \mp \frac{ik_\rho}{k_z} \right. \right. \\
& \left. \left. \times e^{-i\theta} e^{ik_z|z|} \right] \right. \\
& \left. = \frac{ipk_\rho}{4\pi\epsilon_0 k_z} \left[k_0^2 (-i)(\pm) \mathbf{v}_2^\pm + k_\rho k_z (-i)(\pm) \mathbf{v}_1^\pm \right] e^{-i\theta} e^{ik_z|z|} \right.
\end{aligned}$$

Therefore, our expression for the scattered field will be as follows:

$$\mathbf{E}^{\text{scat}}(\rho, \theta, z) = \frac{\pm p}{4\pi\epsilon_0} \int_0^\infty \frac{k_\rho}{k_z} \left[k_0^2 \mathbf{v}_2^\pm + k_\rho k_z \mathbf{v}_1^\pm \right] e^{-i\theta} e^{ik_z|z|} dk_\rho,$$

which is equivalent to **Equation 6** of the paper.

Section 4: Reflected field of the scattered field of the dipole

The reflected field associated with the scattered field may be obtained from calculating the scattered field at $z = -h$:

$$\mathbf{E}^{\text{scat}}(\rho, \theta, -h) = -\frac{p}{4\pi\epsilon_0} \int_0^\infty \frac{k_\rho}{k_z} \left[k_0^2 \mathbf{v}_2^- + k_\rho k_z \mathbf{v}_1^- \right] e^{-i\theta} e^{ik_z h} dk_\rho$$

From this expression, we immediately get the reflected field associated with the scattered field:

$$\mathbf{E}^{\text{scat,ref}}(\rho, \theta, z) = -\frac{p}{4\pi\epsilon_0} \int_0^\infty \frac{k_\rho}{k_z} \left[k_0^2 \mathbf{v}_2^+ + k_\rho k_z \mathbf{v}_1^+ \right] e^{-i\theta} e^{ik_z z} e^{2ik_z h} dk_\rho,$$

which is **Equation 7** of the paper

Section 5: CPA condition

For the first polarization (corresponding to $\mathbf{v}_1^\pm$), the CPA condition reads:

$$\frac{p}{4\pi\epsilon_0} k_\rho^2 (1 - e^{i\delta}) + E_1(k_\rho) e^{i\delta} = 0,$$

which is just obtained by requiring the reflected, scattered-reflected and scattered fields to sum to zero. For the second polarization, the CPA condition reads:

$$\frac{p}{4\pi\epsilon_0} \frac{k_0^2 k_\rho}{k_z} (1 - e^{i\delta}) + E_2(k_\rho) e^{i\delta} = 0$$

These two equations are **Equation 9** and **Equation 10** of the paper, respectively. At this point, all we need to do is find the dipole moment:

$$\begin{aligned} \mathbf{p} &= p e^{-i\theta} \left[\mathbf{e}_\rho - i \mathbf{e}_\theta \right] = \epsilon_0 \alpha_0 \left[- \frac{p}{4\pi\epsilon_0} \int_0^\infty e^{-i\theta} e^{i\delta} \left[\frac{k_0^2 k_\rho + k_z^2 k_\rho}{2k_z} (i \mathbf{e}_\rho + \mathbf{e}_\theta) \right] dk_\rho \right. \\ &\quad \left. - \frac{p}{4\pi\epsilon_0} \int_0^{k_0} \left[\frac{1 - e^{i\delta}}{2e^{i\delta}} \right] \left(k_z k_\rho + k_0^2 \frac{k_\rho}{k_z} \right) \left[e^{i\delta} - 1 \right] (i \mathbf{e}_\rho + \mathbf{e}_\theta) e^{-i\theta} dk_\rho \right], \end{aligned}$$

where we used the expressions for $E_{1,2}(k_\rho)$ in terms of the dipole moment, p . We use the second equality above to write the following equation for the bare polarizability:

$$\frac{1}{\alpha_0} = \frac{1}{8\pi} \left[-i \int_0^\infty e^{i\delta} \frac{k_\rho}{k_z} (k_0^2 + k_z^2) dk_\rho + i \int_0^{k_0} \frac{k_\rho}{k_z} (k_z^2 + k_0^2) \frac{(1 - e^{i\delta})^2}{e^{i\delta}} dk_\rho \right],$$

which is **Equation 11** of the paper (with a difference in minus sign...)

Section 6: Effective polarizability

To find the effective polarizability (**Equation 8** of the paper), we note:

$$\mathbf{p} = \epsilon_0 \alpha_0 \left[\mathbf{E}^{\text{inc}} + \mathbf{E}^{\text{ref}} + \mathbf{E}^{\text{scat, ref}} \right] \rightarrow \mathbf{p} - \epsilon_0 \alpha_0 \mathbf{E}^{\text{scat, ref}} = \epsilon_0 \alpha_0 \left[\mathbf{E}^{\text{inc}} + \mathbf{E}^{\text{ref}} \right]$$

We now use our expression for the reflected scattered field:

$$\mathbf{E}^{\text{scat,ref}}(\rho, \theta, z) = -\frac{p}{4\pi\epsilon_0} \int_0^\infty \frac{k_\rho}{k_z} \left[k_0^2 \mathbf{v}_2^+ + k_\rho k_z \mathbf{v}_1^+ \right] e^{-i\theta} e^{ik_z z} e^{2ik_z h} dk_\rho,$$

From which we obtain:

$$\begin{aligned} \mathbf{E}^{\text{scat,ref}}(0, \theta, 0) &= -\frac{ip}{8\pi\epsilon_0} \int_0^\infty \frac{k_\rho}{k_z} \left[k_0^2 + k_z^2 \right] e^{-i\theta} e^{i\delta} [\mathbf{e}_\rho - i\mathbf{e}_\theta] dk_\rho \rightarrow \\ p \left[1 + \frac{i\alpha_0}{8\pi} \int_0^\infty \frac{k_\rho}{k_z} \left[k_0^2 + k_z^2 \right] e^{i\delta} dk_\rho \right] [\mathbf{e}_\rho - i\mathbf{e}_\theta] e^{-i\theta} &= \epsilon_0 \alpha_0 \left[\mathbf{E}^{\text{inc}} + \mathbf{E}^{\text{ref}} \right] \end{aligned}$$

Thus, we obtain:

$$\mathbf{p} = p [\mathbf{e}_\rho - i\mathbf{e}_\theta] e^{-i\theta} = \epsilon_0 \left[\alpha_0^{-1} + \frac{i}{8\pi} \int_0^\infty \frac{k_\rho}{k_z} \left[k_0^2 + k_z^2 \right] e^{i\delta} dk_\rho \right] \left[\mathbf{E}^{\text{inc}} + \mathbf{E}^{\text{ref}} \right],$$

which is **Equation 8** of the paper (with a difference in minus sign...)